MEWRE Renewable PM: the cheat-sheet pack

Everything from the trial plan on paper: terms, one-line explanations, equations, numbers, actors and what a PM does with them. Built for memorising on a flight. Kuwait facts carry their confidence: H official or signed, M trade press or secondary, L single weak source, V verify before quoting.

17,540 MW
2026 summer peak (MEWRE) M-H

~21 / ~17 GW
installed / reliably available M

15 % by 2030
official renewable target H

2 fils/kWh
citizen tariff since 1966 H

How to study this pack (no copying)

1. **Read one box.** Don't write.
2. **Cover it, say it out loud** in two sentences, as if to a colleague.
3. **Check.** Mark only what you missed (a dot in the margin).
4. **Equations:** say what each letter is and its unit, then one number example.
5. **Next morning:** re-read only the boxes with dots.

25 minutes on, 5 off. Memorise **bold** words and the HOOK lines first; the rest is support.

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The 30 lines to say out loud (if you remember nothing else)

1. **Voltage** is the push, **current** the flow, **resistance** the narrow pipe: V = I R.
2. **MW** is speed, **MWh** is distance: energy = power × time.
3. **DC** pushes one way; **AC** reverses 50 times a second in Kuwait. PV and batteries make DC; inverters make AC.
4. The grid is AC because **transformers** need a changing current to change voltage.
5. Heat lost in wires is **I^2R**: high voltage means low current means low losses. Hence 400 kV.
6. A **coil** resists changes in current, so its current **lags**; a **capacitor** resists changes in voltage, so its current **leads**. ELI the ICE man.
7. **Reactive power (Mvar)** is energy sloshing back and forth each cycle; it does no work but fills cables and moves **voltage**.
8. **$S^2 = P^2 + Q^2$**; power factor = P/S. Equipment is rated in **MVA**.
9. On transmission lines X ≫ R, so **reactive flow sets voltage**, real-power flow sets angle.
10. **Frequency** (50 Hz) is the balance of supply and demand at every instant; **inertia** slows how fast it falls.
11. **Fault level** (MVA or kA) is how much current flows into a short circuit: grid strength and breaker duty.
12. **SCR** = fault level ÷ plant MW. Below about 3 the grid is **weak** for grid-following inverters.
13. Inverters **clamp their current** at about 1.1 to 1.2 pu, so they give little fault current: protection must be checked.
14. **Grid-following** inverters follow the grid's voltage; **grid-forming** ones create it.
15. **EMT studies** simulate instantaneous waveforms in microseconds; needed for weak grids, control interactions and fast transients.
16. **RMS studies** simulate 50 Hz phasors in milliseconds; used for stability over seconds.
17. A PV plant's **MWp** (DC) is bigger than its **MWac**: quote MWac at the peak hour.
18. **Performance ratio** = actual yield ÷ what the sun allowed; about 80 to 85 % for a good plant.
19. **P50** is the median yield, **P90** the yield exceeded 9 years in 10: lenders size debt on P90.
20. **Battery MW** is the tap, **MWh** the tank; hours = MWh ÷ MW.
21. The **critical path** has zero float: any slip on it moves the end date.
22. **CPI** = EV/AC (cost), **SPI** = EV/PV (schedule); below 1 is bad.
23. **FIDIC Silver** (EPC) puts most risk on the contractor; **Red** keeps design with the employer.
24. A claim needs a **notice within 28 days** or it is time-barred (FIDIC 2017, 20.2.1).
25. **LDs** are agreed damages per day of delay, capped (typically 10 to 20 %).
26. **LCOE** = lifetime discounted cost ÷ discounted MWh. Gulf solar tariffs are about 1.0 to 1.7 US¢/kWh.
27. **DSCR** = cash available ÷ debt service; lenders want about 1.2 to 1.3×.
28. IPPs go **MEWRE → KAPP → Higher Committee → bids → Audit Bureau → PPA → financial close**.
29. Tenders above **KD 75,000** go through **CAPT**; Audit Bureau prior review from about KD 100,000 V.
30. Always say a number **with its source and date**; never call a G2G deal "contracted" before signature.

1 · Kuwait: numbers to know cold

source: MEWRE track modules 0 and 11, trial playbook

Say a number with its source and confidence. Exchange rate assumed 1 KWD ≈ 3.25 USD; 1 KWD = 1,000 fils. Fiscal year 1 April to 31 March H.

System

Item	Value	
Peak demand 2024 / 2025 / 2026	17,640 / 17,610 / **17,540 MW**	M-H
Record monthly peak (MEES basis)	17.71 GW, July 2025	M
June 2025 peak vs available	16.98 vs 16.275 GW → reserve margin ≈ −4.2 %	M
Installed / reliably available / gas	~21 / ~17 / ~14 GW	M
Generation mix	~62 % gas, ~36 % oil, ~2 % renewable	M-L
New connections	~1 GW per year	M
Planned additions	14.05 GW by 2031	M
GCC interconnection	500 MW deal (2024); KWD 161.381 m approved by Audit Bureau 2025 (conditional)	M
Frequency	50 Hz	H

HOOK Summer peak ≈ 17.5 GW, reliably available ≈ 17 GW: Kuwait is short in summer. Every summer MW matters.

Power cuts to know

- **Jun 2024:** 300 MW Az-Zour South trip.
- **Aug 2024:** gas supply shortfall.
- **Apr 2025:** heat plus maintenance.
- **May 2025:** more than 1.5 GW short.

IN PRACTICE Plan grid outages for connection works in **November to March**, never in summer.

Targets

Official renewable share	**15 % by 2030**	H
Stated MEWRE ambition	30 / 40 / 50 % by 2030 / 2040 / 2050	M
Net zero	2060 national; 2050 oil and gas sector	H
NDC	−7.4 % vs 2035 business as usual	H
Rystad forecast	~3.3 GW (~7 %) renewables by 2030	M

$P_{RE} = \sigma \cdot E_{total} / (CF \cdot 8{,}760)$ capacity needed for share σ

15 % of ~84 TWh at CF 25 % ≈ 5.75 GW (illustrative, track example).

Pipeline

Project	Status	
Shagaya Phase I, 70 MW	50 MW CSP + 10 MW PV + 10 MW wind; operating since 2017 to 2019; run by KISR; CSP data after 2019 not public	H / L
Zone 1 PV, 1,100 MW (Shagaya Phase III)	RFQ Jan 2024 → bids 15 Jan 2026 → Masdar + Fouad AlGhanim reported selected ~Aug 2026; tariff not published; own 400 kV substation	M
Zone 2 PV, 500 MW	RFQ May 2025 → RFP Nov 2025 → bids 26 Jul 2026 (2 or 3 bids)	M
400 kV Shagaya to Al-Wafra line	Awarded to Power Grid Co., KD 48.6 m, Mar 2026 (≈ 11 months bid to award)	M
BESS programme	1 to 1.5 GW / 4 to 6 GWh; first 500 MW for summer 2027	M-L
China G2G Phases 3 to 4	NEA framework 17 Mar 2025; SPIC; 4.8 GW claimed: **not "contracted"**	L-M
Az-Zour North 2 and 3 IWPP	ECWPA signed 3 Feb 2026; ACWA + GIC 40 % / KAPP 60 %	H
Al-Khairan Phase 1 IWPP, 1,800 MW	**Gas-fired** water and power; 2 bids 1 Jun 2026 (no 1,800 MW Al-Khairan solar found)	H
Dibdibah, Abdaliya	Cancelled	M

Tariffs and the value of a solar MWh

Citizens (private housing)	**2 fils/kWh** since 1966	H
Apartments	5 / 10 / 15 fils/kWh (2016 law, from Aug 2017)	H
Commercial / industrial-agricultural	5 / 3 fils/kWh	H
Cost of supply	14 to 38 fils/kWh (no official current figure)	M
Fuel displaced by solar	≈ 26 (LNG) to 37 (crude) fils/kWh	M
Solar tariff (analyst estimate)	7 to 8 fils/kWh; Zone 1 not reported	L
Cost of delay, 1.1 GW plant	≈ KWD 191k per day	M

Subsidy per kWh s = cost − tariff · Value of displaced fuel v = heat rate × fuel price

Cost of delay $C_d = \Delta t \cdot P \cdot CF \cdot 24 \cdot (v - T)$

HOOK Every solar kWh at ~8 fils saves fuel worth ~26 to 37 fils that Kuwait could sell: that is the national case for renewables.

Budget (FY26/27 draft)

Spending KD 26.1 bn, revenue KD 16.3 bn, capex KD 3.1 bn M. Submit budget requests in the right fiscal year (1 Apr to 31 Mar).

2 · Who decides what: the actors

modules 0, 11, 13

The institutions

MEWRE وزارة الكهرباء والماء والطاقة المتجددة	Ministry of Electricity, Water and Renewable Energy (est. 1962). **Policy, grid owner and operator, offtaker** (30-year solar PPAs; 25-year ECWPA for water and power), tenders grid works.
KAPP هيئة مشروعات الشراكة بين القطاعين العام والخاص	Kuwait Authority for Partnership Projects (PPP Law 116/2014, Decree 78/2015, attached to Ministry of Finance). **Procures PPP/IPP projects**; holds 60 % of IWPP project companies in practice.
Higher Committee اللجنة العليا لمشروعات الشراكة	Chaired by the **Minister of Finance**. Approves the project, its stages and the selected investor.
CAPT الجهاز المركزي للمناقصات العامة	Central Agency for Public Tenders (Law 49/2016, amended by Law 94/2026). **State-funded tenders above KD 75,000** (EPC route).
Audit Bureau ديوان المحاسبة	State Audit Bureau (Law 30/1964). **Prior review** of large contracts, reported from KD 100,000 V. Outcomes: approve, reject or conditional.
Fatwa & Legislation إدارة الفتوى والتشريع	Legal review of contract terms (stage in the chain unconfirmed).
KISR معهد الكويت للأبحاث العلمية	Kuwait Institute for Scientific Research. Runs **Shagaya Phase I**; solar and wind resource measurement.
SCPD المجلس الأعلى للتخطيط والتنمية	Supreme Council for Planning and Development: Vision 2035, development plan.
National Assembly مجلس الأمة	Suspended 10 May 2024: laws are issued by **decree-law**.
KDIPA هيئة تشجيع الاستثمار المباشر	Direct investment promotion: up to 100 % foreign ownership, tax holiday up to 10 years.
KPC مؤسسة البترول الكويتية	Kuwait Petroleum Corporation: pays ~25 fils/kWh (a transfer, not a national saving); under Law 94/2026 under CAPT only above KD 20 m.
GCCIA هيئة الربط الكهربائي الخليجي	GCC Interconnection Authority: 400 kV regional grid; HVDC back-to-back to Saudi Arabia's 60 Hz system.

The IPP approval chain (memorise the order)

1 Request (MEWRE · الوزارة) → 2 Feasibility (KAPP · هيئة الشراكة) → 3 Higher Committee · اللجنة العليا → 4 RFQ → 5 RFP: technical, then financial → 6 Preferred bidder → 7 Higher Committee · اللجنة العليا → 8 Audit Bureau · ديوان المحاسبة → 9 PPA signed · اتفاقية شراء الطاقة → 10 Financial close · الإغلاق المالي

> **HOOK** "Me-Ka-Hi, Q-P-Pre, Hi-Au, PPA-Close": MEWRE asks, KAPP studies, Higher Committee blesses, bids, Higher Committee again, Audit Bureau, sign, money.

Three delivery routes: EPC (state capex, CAPT) · PPP/IPP (KAPP, PPA, private finance) · G2G (government to government).

PPP shareholding (legal guide): 50 % IPO to citizens / 6 to 24 % public bodies / ≥ 26 % private investor, for projects above KD 60 m V. IWPP practice: 40 % developer, 60 % KAPP.

Benchmark	Elapsed
Zone 1 RFQ → reported selection	~31 months
AZN 2&3 letter agreement → ECWPA	~6 months
Al-Khairan RFP → bids	9 months
400 kV line bid → award	~11 months

Inside MEWRE: how authority flows

Level	Arabic	What the PM needs
Minister	الوزير (معالي الوزير)	Reached via the Undersecretary
Undersecretary	وكيل الوزارة	Decision memos, senior external letters
Assistant Undersecretary (sector)	الوكيل المساعد	Your internal client; budget owner
Director / Controller / Section head	مدير إدارة / مراقب / رئيس قسم	Review, drafting, execution

- **Formal authority** = signature + delegation (تفويض الصلاحيات). **Practical authority** = drafters, gatekeepers, trusted advisers.
- The annotation on a memo, the / للعرض / (موافق) **تأشيرة** (لاتخاذ اللازم), **is** the decision. A verbal "go ahead" is agreement in principle only.
- **Consultants** (owner's engineer) advise and prepare; **officials decide and sign**.
- **Wasta test:** relationships may buy access and attention, never outcomes the rules decide. Decline politely, disclose the approach.
- Approval cycle example: ~46 working days for a memo chain; plan for it.

Expected approval time with returns $E[T] = \Sigma\, t_i / (1 - p_i)$ p = chance a step sends the file back

IN PRACTICE Complete files reduce p: the cheapest way to speed approvals is a file that is not returned.

Key laws and thresholds

Instrument	Key content
Tenders Law 49/2016	CAPT over KD 75,000; final guarantee ≥ 10 % (reported)
Law 94/2026	Gazette 27 Sep 2026; effective ~27 Dec 2026. Direct contracting up to **KD 1 m per FY** without CAPT; **agents banned**; ≥ 30 % local products and ≥ 30 % local subcontracting; SME 10 % margin; anti-splitting
Law 1/2024	Registered foreign firms may bid without a local agent
Law 74/2019	National product preference (rate unverified)
PPP Law 116/2014 + Decree 78/2015	KAPP, Higher Committee
Law 30/1964	Audit Bureau prior review

TRAP During the trial week the **old KD 75,000 rule still applies**; Law 94/2026 takes effect around late December 2026.

3 · Electricity from zero

trial Day 1

One picture for everything: a closed water loop. Pump = voltage, flow = current, narrow pipe = resistance, heavy wheel = coil, rubber membrane = capacitor.

Voltage, current, resistance

Charge	The "stuff" that moves (coulombs), carried by electrons already in the wire.
Voltage V	The **push** between two points (volts). Always measured **between** two points.
Current I	The **flow** of charge (amps). Same at every point of a simple loop.
Resistance R	How hard it is to push current (ohms). Turns electrical energy into **heat**.

$V = I \cdot R$ $P_{heat} = I^2 \cdot R$ double the current → four times the heat

Example: 240 V across a 24 Ω heater → 10 A; across 48 Ω → 5 A.

HOOK Push, flow, narrow pipe. Cut the wire: current stops everywhere, but the push (voltage) across the gap stays.

Power vs energy

Power P	Rate of energy flow now: W, kW, **MW**, GW. Plants are sized in MW.
Energy E	Power × time: Wh, kWh, **MWh**, GWh, TWh. Tariffs are per MWh.
Capacity factor	Energy produced ÷ energy at full power all year.

$P = V \cdot I$ $E = P \cdot t$ $CF = E / (P_{rated} \cdot 8{,}760\ h)$

1 MW for 1 h = 1 MWh. 1 MWh = 3.6 GJ. A 1,100 MW PV plant at CF 26 % → 2.5 TWh/yr. Typical CF: PV 20 to 32 %, wind 25 to 45 %, CSP with storage 35 to 55 %, gas 50 to 90 %.

HOOK MW is the speedometer, MWh is the odometer.

Efficiency from heat rate $\eta = 3{,}600 / HR$ HR in kJ/kWh; 9,000 kJ/kWh → 40 %

DC and AC

DC	Push always the same way. Batteries, **PV modules**, electronics inside devices.
AC	Push reverses: **50 Hz** in Kuwait = 50 full back-and-forth cycles a second (period 20 ms). Electrons wiggle; energy travels.
Why AC	**Transformers** only work with a changing current, so AC can be stepped up to 400 kV and down to 240 V cheaply.
RMS	The DC value that heats a kettle equally. Every AC number you hear is RMS.
Inverter	DC → AC (PV, batteries). **Rectifier**: AC → DC (phone charger). **HVDC**: DC links for very long or undersea lines and between grids of different frequency.

$V_{peak} = \sqrt{2} \cdot V_{RMS}$ 240 V RMS peaks at 339 V $\omega = 2\pi f = 314$ rad/s

HOOK A hand saw cuts on both strokes without travelling: that is AC.

Magnetism, induction, coils

- **Current makes a magnetic field.** An iron core concentrates it.
- **A changing field makes a voltage** (Faraday's law of induction). Steady field → nothing.
- **Generator:** spinning magnet past coils; 2-pole machine at 3,000 rpm = 50 Hz.
- **Transformer:** AC in one coil → changing flux in the core → voltage in the second coil.
- **Motor:** current in a magnetic field feels a force.
- **Inductance L** (henry): a coil resists **changes** in current (heavy wheel). In AC its current **lags** the voltage by up to 90° (5 ms). Stores energy in its magnetic field.

Capacitors

- Two plates and a gap; stores charge in an **electric field** (farads).
- Resists **changes in voltage** (rubber membrane). **Blocks DC** once charged.
- In AC its current **leads** the voltage by up to 90°.
- In the grid: **capacitor banks** (power-factor correction), **long cables** behave like capacitors (charging current, ~1 to 2 Mvar/km at 132 kV, ~10 Mvar/km at 400 kV, typical), inverter DC links.

HOOK ELI the ICE man: in an inductor (L) voltage E leads current I; in a capacitor (C) current I leads voltage E.

Reactance and impedance

Reactance X	Opposition of coils and capacitors to AC (ohms). **Stores and returns** energy instead of burning it.
Impedance Z	Resistance and reactance combined: sets the AC current. Adds **at right angles**.
Resonance	$X_L = X_C$: they cancel; only R limits current. Source of harmonic problems.

$X_L = 2\pi fL$ $X_C = 1/(2\pi fC)$ $Z = \sqrt{(R^2 + X^2)}$ $\varphi = \arctan(X/R)$

R = 30 Ω, X = 40 Ω → Z = 50 Ω (not 70), I = 4.8 A at 240 V, φ = 53°, PF = 0.6.

Overhead lines are long thin coils: at 400 kV X is about **10 to 40 × R** (≈ 0.28 to 0.33 Ω/km).

Per unit (pu)

Express quantities as a fraction of a base (e.g. 100 MVA, 400 kV) so transformers disappear from calculations. 1.05 pu voltage = 5 % above nominal.

$Z_{base} = V_{base}^2 / S_{base}$ $z_{pu} = Z / Z_{base}$

4 · Grid electricity

trial Day 1 and 3, module 1 and 6

Real, reactive, apparent power

P (MW)	Real power: does work (the beer).
Q (Mvar)	Reactive power: energy sloshing each cycle (the foam). Lagging loads (motors, AC compressors) **absorb** Q; capacitors and cables **supply** it.
S (MVA)	Apparent power: what cables and transformers must carry (the whole glass).
Power factor	P/S = cos φ. AC loads ~0.8 to 0.9 lagging.

$S^2 = P^2 + Q^2$ $P = V\,I\cos\varphi$ $Q = V\,I\sin\varphi$ $PF = P/S$

$Q = P \cdot \tan(\arccos PF)$ at PF 0.95, Q = 0.329 P

Capacitor for PF correction $Q_c = P(\tan\varphi_1 - \tan\varphi_2)$

Only ~3 % of each cycle's energy flows back at PF 0.8, but the current is 25 % higher: the cost of Q is **current** (capacity, losses, voltage drop).

Why reactive power moves voltage

$\Delta V \approx (R \cdot P + X \cdot Q) / V$ (pu: $\Delta v \approx r\,\Delta p + x\,\Delta q$, $x \approx S_{base}/S_{sc}$)

Since X ≫ R on HV lines, **Q flow sets voltage**; P flow sets the angle between buses. Fix low voltage by supplying Q **near the load**: capacitor banks, SVC, **STATCOM**, PV and BESS inverters, generators.

IN PRACTICE Grid codes ask PV plants for Q at the point of interconnection, typically about **0.95 lagging to 0.95 leading** at full output across 0.90 to 1.10 pu voltage. Kuwait's code text was not found: ask MEWRE Transmission.

Worked: 1,100 MW at PF 0.95 needs 361.5 Mvar at the POI → inverters ≈ 584 Mvar / ~1,260 MVA incl. transformer losses; with 8 % inverter derating at 50 °C → ~308 Mvar short → STATCOM or oversizing.

Three-phase

- Three sine waves 120° apart. Total power is **constant** (single phase pulses 0 to 2× average).
- Balanced currents cancel: 3 wires instead of 6. Creates a **rotating magnetic field** for motors.
- **Line (phase-to-phase)** vs **phase (to neutral)** voltage: 415 V / 240 V; 33 kV / 19.05 kV.
- **Star (wye)** has a neutral; **delta** doesn't. Vector groups (Dyn11, YNd11) describe winding connections and phase shift.
- **Unbalance:** unequal phase loads → neutral current, negative sequence, motor heating.

$V_L = \sqrt{3} \cdot V_{ph}$ $S = \sqrt{3}\, V_L\, I_L$ $P = \sqrt{3}\, V_L\, I_L \cos\varphi$

30 MW at PF 0.95 on 33 kV: 31.6 MVA ÷ (1.732 × 33 kV) = 553 A. 400 kV × 2.5 kA × √3 = 1,732 MVA.

Transformers and voltage levels

$V_1/V_2 = N_1/N_2 = I_2/I_1$ losses $\propto I^2R \propto 1/V^2$

- Moving 1,100 MW at 400 kV instead of 132 kV: current ÷ 3, losses ÷ ~9.
- **% impedance** (z): voltage drop on load and the fault current let through. $I_{sc} \approx I_{rated} / z$ (z in pu).
- **OLTC** (on-load tap changer): adjusts ratio in ~1.25 % steps to hold voltage.
- **Cooling**: ONAN (oil natural, air natural), ONAF (fans), OFAF. Rating at **50 °C** ambient matters in Kuwait (rough rule ~1 %/°C derating above design ambient; manufacturer to confirm).
- **Inrush**: energising a transformer draws a large magnetising current spike for seconds.

Kuwait levels: 400 kV confirmed; older EHV quoted 300 or 275 kV V; 132, 33, 11 kV and 415/240 V commonly cited M.

Frequency, inertia, response

Frequency	50 Hz = supply exactly equals demand. Generation trip → frequency falls.
Inertia H	Stored rotating energy (seconds of rated power). Slows the fall. PV and BESS have none physically.
RoCoF	Rate of change of frequency (Hz/s) right after a trip.
Droop	Primary response: output rises in proportion to the frequency drop (e.g. 4 %: full output change for a 4 % = 2 Hz drop).
UFLS	Under-frequency load shedding: the last resort.

$RoCoF = \Delta P \cdot f_0 / (2\, H\, S_{sys})$ 300 MW trip, H 4 s, 15 GVA → 0.125 Hz/s

$Droop\ \Delta P = -(P_n/R) \cdot (\Delta f/f_n)$ 500 MW BESS, 4 %, at 49.6 Hz with ±0.2 Hz deadband → 50 MW

Layer	Time
Inertia (+ BESS fast response < 0.5 to 1 s)	0 to 5 s
Primary / droop	2 to 30 s
Secondary / AGC	30 s to 15 min
Tertiary / reserves	≥ 15 min

Fault level and protection

Short circuit	A low-impedance path: current jumps to many times normal until a breaker clears it.
Fault level	Current (kA) or MVA that flows into a fault at a bus: measures grid **strength** and sets **breaker duty**.
X/R ratio	Higher X/R → larger DC offset in fault current → harder for breakers (transmission X/R typically 10 to 20+).
Protection	CTs/VTs → relay (IED) → breaker trip. 80 to 120 ms total clearance at 400 kV.

$S_{sc} = \sqrt{3} \cdot V \cdot I_{sc}$ $S_{sc} = S_b / (z_s + z_T)$, $z_s = S_b/S_{sc,source}$

Worked: 300 MVA, 14 % transformer behind a 15 GVA 400 kV bus → 32.8 kA at 33 kV > 31.5 kA switchgear: under-rated. Two in parallel → 58 kA: keep the 33 kV coupler normally open.

ANSI code	Function
21	Distance (line)
87L / 87T / 87B	Differential: line / transformer / busbar
50 / 51	Instantaneous / time overcurrent
50BF	Breaker failure
25	Synchro-check
27 / 59	Under / over-voltage
81	Frequency

Relay setting ohms $Z_{sec} = Z_{prim} \cdot CTR / VTR$ zone 1 ≈ 0.8 Z_{line}, zone 2 ≈ 1.2 Z_{line}

Kuwait's load curve

- **Summer afternoon peak** driven by air conditioning (~17.5 GW); winter far lower.
- Load factor ≈ 55 % (84 TWh ÷ 17.5 GW ÷ 8,760, illustrative).
- PV output peaks at noon, demand stays high into the evening → **net-load peak moves to ~18:00** (duck curve) → batteries or CSP storage for the evening.

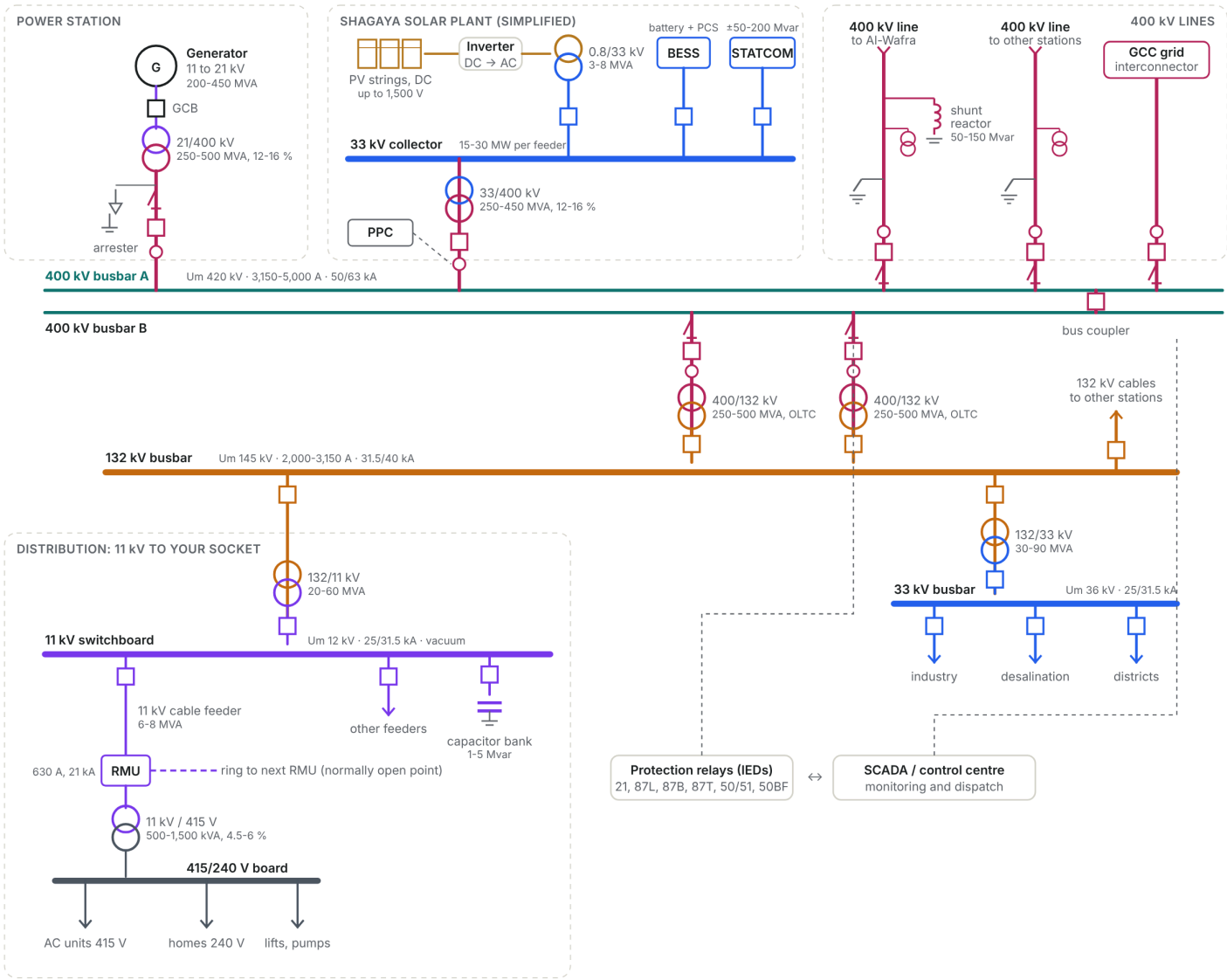
> Load factor $LF = E / (P_{peak} \cdot 8{,}760)$ Reserve margin $RM = (C_{avail} - P_{peak}) / P_{peak}$

Substation essentials

Bay	One circuit's set: disconnectors, breaker, CTs, VT, earthing switches, arresters.
Busbar schemes	Single bus · double bus + coupler · ring · **breaker-and-a-half (EHV standard)** · double breaker.
AIS / GIS	Air-insulated (cheap, large, dust and pollution flashover risk) vs gas-insulated (compact, sealed against dust, SF6 management, vendor lock-in).
Disconnector	Visible isolation; **never** breaks load.
Shunt reactor	Absorbs Q from long lines and cables at light load.

5 · Kuwait grid single-line diagram and typical ratings

simplified teaching drawing



Read the drawing top to bottom

1. **Power station:** generator 11 to 21 kV → generator breaker → **step-up transformer** 21/400 kV → 400 kV.
2. **Solar plant:** PV DC (≤ 1,500 V) → inverter 0.6 to 0.8 kV AC → 0.8/33 kV transformer → **33 kV collector** (with BESS, STATCOM) → **33/400 kV plant transformer** → point of interconnection (POI). The **PPC** controls it all at the POI.
3. **400 kV double busbar** with bus coupler; lines to Al-Wafra and other stations (shunt reactor); **GCC interconnector**.
4. **400/132 kV** transformers (two or more for N-1) → 132 kV busbar and cables.
5. **132/33 kV** → industry, desalination, districts. **132/11 kV** → 11 kV street network.
6. 11 kV feeder → **RMU** (rings, normally-open point) → **11 kV/415 V** distribution transformer → 415 V (AC units) / 240 V (sockets).
7. **Every bay:** disconnector → breaker → CT (+ VT, earthing switch, surge arrester). IEDs protect, SCADA reports to the control centre.

Typical ratings by voltage level (IEC values + common practice)

System	Um	BIL	Current	Breaker kA	Transformers
400 kV	420 kV	1,425 kV	3,150 to 5,000 A	50 / 63	GSU, plant, 400/132: 250 to 500 MVA, z 12 to 16 %
132 kV	145 kV	650 kV	2,000 to 3,150 A	31.5 / 40	132/33: 30 to 90 MVA; 132/11: 20 to 60 MVA
33 kV	36 kV	170 kV	1,250 to 2,500 A	25 / 31.5	inverter transformers 3 to 8 MVA
11 kV	12 kV	75 kV	630 to 2,500 A	25 / 31.5	distribution 500 to 1,500 kVA, z 4.5 to 6 %
415/240 V	< 1 kV	n/a	boards ≤ 4,000 A	25 to 50	1,000 kVA → ~1,390 A on LV

Um = highest equipment voltage (a "400 kV" breaker is built for 420 kV). **BIL** = lightning impulse withstand. Breaker kA must exceed the study's fault current.

Equipment	Typical
CT	1 A secondary (HV); 0.2S metering; 5P20 / PX / TPY protection
VT / CVT	400/√3 kV : 110/√3 V; 0.2 metering, 3P protection
Surge arrester (420 kV system)	U_r ≈ 336 to 360 kV, 10 to 20 kA, metal-oxide
Shunt reactor 400 kV	50 to 150 Mvar
400 kV line	~1,500 to 2,500 MVA per circuit (summer), SIL ~500 to 600 MW
RMU	630 A ring switches, 20 to 21 kA
Inverter	central 3 to 5 MVA, string 200 to 350 kVA, ~98.5 to 99 %
STATCOM	±50 to ±200 Mvar, < 50 ms
BESS	3 to 5 MWh per container; PCS 2.5 to 4 MVA
Protection	relay 20 to 30 ms; total clearance 80 to 120 ms at 400 kV; breaker failure 150 to 250 ms
Metering at POI	0.2S meter + 0.2S CT + 0.2 VT; main and check meters

6 · Inverters on the grid: SCR, clamping, ride-through, control

modules 6, trial Day 3

Inverter-based resources (IBRs) = PV, BESS, wind converters, STATCOMs, HVDC. They behave very differently from spinning generators: fast, controllable, but with limited current and no natural inertia.

The SCR family (grid strength)

SCR	**Short-circuit ratio** = fault level at the POI ÷ plant rating. How "stiff" the grid is relative to the plant.
WSCR	Weighted SCR: fault level ÷ **sum** of all nearby IBR ratings (plants interact through the same weak node).
ESCR	Effective SCR: subtracts capacitor banks and filters at the POI, which make the grid look weaker: $(S_{sc} - Q_c)/P$.
CSCR	Composite SCR: an equivalent for a whole area with many IBRs.
System strength	The grid's ability to hold voltage waveform and angle stable when disturbed; low fault level = weak.

$SCR = S_{sc,POI} / P_{plant}$ $WSCR = S_{sc} / \Sigma P_{IBR}$

SCR	Meaning (rule of thumb)
> 5	Strong; RMS studies usually enough
3 to 5	Moderate; **EMT required if SCR < ~5** (module 6 rule)
< 3	**Weak**; grid-following inverters struggle (control instability, voltage oscillation)
< 2	Very weak; needs grid-forming, synchronous condensers or network reinforcement

Example: 15,000 MVA at the 400 kV POI, 1,100 MW plant → SCR ≈ 13.6 (strong). Same plant at a 3,000 MVA bus → 2.7 (weak).

ASK What is the minimum fault level at the POI (N-1, minimum generation, winter night), and what SCR does the inverter vendor guarantee stable operation down to?

Grid-following vs grid-forming

Grid-following (GFL)	Measures grid voltage with a **PLL** (phase-locked loop) and injects current in step with it: a **controlled current source**. Needs a stiff grid (SCR ≳ 3). Today's standard PV inverter.
Grid-forming (GFM)	Creates its own voltage waveform: a **controlled voltage source** behind an impedance. Gives synthetic inertia, stabilises weak grids, can black-start. Needs headroom (usually a battery).
PLL	The control that tracks the grid's angle and frequency; in weak grids it can lose lock or oscillate.
Synthetic inertia / fast frequency response (FFR)	Inverter output injected within < 1 s on a frequency drop.
Synchronous condenser	A spinning machine with no prime mover: adds inertia, fault level and Q.

IN PRACTICE Procurement rule from module 6: **require GFM-capable BESS PCS** in specifications, even if operated grid-following at first.

Inverter current clamping (current limiting)

What: an inverter's power semiconductors (IGBTs) are destroyed by overcurrent in microseconds, so the control **clamps (limits) output current** at about **1.0 to 1.2 pu** of rating. A synchronous generator can give 4 to 7 × rated current into a fault; an inverter gives about **1.1 to 1.2 ×**.

Consequences:

- **Low fault current** from IBR-heavy areas: overcurrent and distance relays may not see faults or may misjudge direction. Protection must be studied with IBR models.
- **Priority choice** during a dip: **Q-priority** (reactive current first, to support voltage; usual grid-code requirement) or P-priority.
- **Control saturation**: when clamped, controllers must not "wind up"; poor anti-windup → oscillations after the fault clears.
- **Grid-forming + clamping**: when a GFM inverter hits its current limit it behaves like a current source and can lose synchronism; the vendor's current-limit design is a key study item.
- **Negative-sequence current** during unbalanced faults depends on vendor control: affects protection.

Related "clamps":

DC-link voltage clamp (chopper)	A braking resistor that dumps energy when the inverter can't export during a fault, keeping DC voltage safe. PV can instead move off its maximum power point.
Clipping	Not a fault behaviour: when DC power exceeds the inverter's AC rating (DC/AC ratio > 1), output is **flat-topped** at the AC limit at midday.
Surge arrester	"Clamps" overvoltage spikes to earth (insulation protection).

HOOK A generator is a heavy truck that can push through a wall; an inverter is a sports car with a rev limiter: fast and precise, but it will not give more than ~1.2 × its rating.

ASK What is the inverter's maximum fault current, for how long, and is it reactive-priority? Is the current-limit behaviour in the EMT model the real firmware?

Fault ride-through (FRT)

LVRT / HVRT	Stay connected through low / high voltage dips and swells inside a voltage-time envelope instead of tripping.
Envelope (comparator)	Ride through 0 pu for ~150 ms, then recover along a ramp (e.g. to 0.85 pu by 1.5 s); HVRT to ~1.2 to 1.3 pu. Kuwait values: from MEWRE's code.
Reactive current injection	During the dip, inject reactive current in proportion to the voltage drop.
Active power recovery	Return to pre-fault P within a set time after clearance (too fast → voltage problems; too slow → frequency dip).
Momentary cessation	Inverter stops injecting during a dip; widely restricted after US events (Blue Cut 2016, Odessa 2021).

$\Delta I_q = k \cdot \Delta V$ k typically about 2 (range 2 to 6 by code), in pu

HOOK Why it matters: one fault can trip hundreds of MW of solar at once if FRT settings are wrong, turning a small fault into a frequency event.

Power plant controller (PPC) modes

P control	Follows active-power set-point or curtailment order; **ramp-rate limit** (≤ ~10 %/min, comparator).
Q control	Fixed Mvar at the POI.
PF control	Fixed power factor at the POI.
Voltage control (with droop/slope)	Q changes with voltage error (e.g. 2 to 5 % slope); preferred by grid operators.
Frequency response	**LFSM-O**: reduce P on over-frequency; **LFSM-U**: raise P on under-frequency (needs headroom); **FSM**: continuous droop.

Response times: inverter FRT milliseconds; PPC loop 100 to 200 ms; voltage control 1 to 5 s; set-point changes ~1 min.

Power quality

Harmonics	Currents at multiples of 50 Hz from power electronics. **THD** = total harmonic distortion. IEEE 519: 1.5 % THD above 161 kV.
Resonance	Cables and capacitors with grid inductance amplify certain harmonics; found by frequency scans.
Flicker	Fast voltage fluctuation visible in lighting (P_{st}, P_{lt}); from cloud transients or wind.
Unbalance	Negative-sequence voltage, typically limited to ~1 to 2 %.
Rapid voltage change	Step on switching capacitor banks or energising transformers.

$$THD = \sqrt{(\Sigma\ V_h^2)} / V_1$$

Worked: harmonic study result THD 1.63 % > 1.5 % limit → filter or resonance mitigation.

Curtailment and deemed energy

- **Causes:** grid constraint, system security, oversupply, late or unavailable grid (offtaker risk → **deemed energy**), plant faults (IPP risk).
- **Define in the PPA:** how available power is estimated (reference inverters), data access, logging with cause codes, allowance before payment.

$$\text{Deemed energy} = \Sigma\ (P_{avail} - P_{actual})\ \Delta t;\ \text{pay}\ \max(0,\ E - E_{allow}) \cdot T$$

7 · Power system studies: what each answers

module 6, trial Day 3

1 Load flow (power flow)

Answers: do lines and transformers overload, do voltages stay in limits, in normal (N) and single-contingency (N-1) cases?

Cases for Kuwait: summer peak, **winter midday** (low load, high solar: overvoltage, reverse flow), evening BESS discharge, N-1 outages.

Example: one double-circuit 400 kV line (2 × 1,732 MVA) + 1.6 GW → OK under N-1 (97 %); a further 1.7 GW → fails N-1.

2 Short circuit (IEC 60909)

Answers: maximum fault currents (breaker and busbar ratings, earthing) and **minimum** fault levels (SCR, protection sensitivity). Uses voltage factor c (1.1 for max at HV).

3 RMS (phasor) stability

Model: positive-sequence 50 Hz phasors; time step ~1 to 10 ms; simulates seconds to minutes.

Answers: transient (angle) stability after faults, voltage stability, frequency response after trips, PPC and FRT behaviour at system level.

Tools: PSS/E, PowerFactory (RMS), PSLF.

4 EMT (electromagnetic transients)

Model: instantaneous three-phase **waveforms**, time step **~1 to 50 µs**, detailed inverter controls (ideally the vendor's real firmware code), switching, saturation. Slow: usually a local area, seconds of time.

Needed when:

- **Weak grid** (SCR < ~3 to 5): GFL inverter control stability, PLL.
- **Control interactions** between nearby IBRs, STATCOMs, HVDC (oscillations, including sub-synchronous control interaction).
- **Fault ride-through** verification with real current limiting and unbalanced faults.
- **Grid-forming** behaviour and current-limit transitions.
- **Energisation:** transformer inrush and sympathetic inrush, cable and capacitor switching.
- **Temporary overvoltage (TOV)** after load rejection or fault clearing; ferroresonance.
- **Harmonic resonance** and protection performance with IBR fault currents.

Tools: PSCAD/EMTDC, EMTP, PowerFactory EMT, RTDS (real-time, hardware-in-the-loop).

> **HOOK** RMS = a movie at 100 frames per second showing the smooth average; EMT = a slow-motion camera at 50,000 frames per second showing every wiggle.

5 Harmonics and resonance

Frequency scans of the network impedance plus inverter harmonic emissions → THD vs limits; filter design.

6 Protection coordination

Relay settings grade so the nearest breaker trips first; checks with low IBR fault current; settings approved by MEWRE before energisation.

7 Insulation coordination

Matches equipment insulation (BIL, switching withstand) to lightning, switching and temporary overvoltages; places surge arresters. Uses EMT for switching surges.

Other studies you will hear about

Earthing (grounding)	Step and touch voltages at substations (IEEE 80).
Arc flash	Incident energy for worker PPE.
Cable sizing / ampacity	Current rating in hot desert soil.
Reactive capability	P-Q diagram at the POI across voltage and 50 °C.
Grid impact / interconnection study	The package of the above for a connection application, done **cumulatively** with neighbouring plants.

8 · Managing studies and grid compliance in practice

what the PM actually does

Study management sequence

1. **Scope agreement** with MEWRE Transmission: study list, cases (summer peak, winter midday, N-1), SCR range, future projects included, criteria, deliverables.
2. **Network data:** MEWRE issues the latest network model (PSS/E / PowerFactory) under NDA; agree the base year.
3. **Plant models:** vendor RMS and EMT models, version-locked to the firmware; parameter sheets; model validation report.
4. **Model acceptance tests:** initialises cleanly, flat run (no drift), step responses, FRT, RMS vs EMT benchmark.
5. **Run studies** (consultant or EPC), cumulative with neighbouring plants.
6. **Review** with the owner's engineer against the checklist; comments and resubmission.
7. **Approval** → settings and design frozen → equipment orders.
8. **Compliance tests** on site match the study: the model must match the field.
9. **As-built model** handed over after commissioning; re-study if firmware changes.

Study review checklist

- Right network base case and year? Future plants included?
- Minimum and maximum fault levels both used?
- Worst cases: winter midday, N-1, low inertia?
- Plant model = the actual inverter, PPC and firmware version ordered?
- EMT used where SCR is low or near other IBRs?
- Reactive capability at 50 °C and 0.90 to 1.10 pu?
- FRT, reactive current injection and recovery shown against the code?
- Harmonic limits met at the POI with background distortion?
- Protection settings consistent with low IBR fault current?
- Assumptions, limitations and sensitivities stated?

Model requirements to write into the contract

- Site-specific **RMS and EMT models** of inverter, PPC, STATCOM, BESS, delivered early and updated.
- EMT model uses the **real control code** (not a generic approximation); black-box allowed under NDA.
- **Validation** against type tests, factory tests or hardware-in-the-loop; then against site tests.
- **Firmware change control:** any firmware or setting change needs notification, updated models and possibly re-study.
- Model documentation: parameters, valid range (SCR, X/R), time step.

Grid code compliance checklist

- **Comparators** if Kuwait's code text is unavailable: ENTSO-E RfG, IEEE 2800, Saudi and Abu Dhabi codes, GCCIA.
- Frequency and voltage operating ranges; RoCoF withstand.
- FRT (LVRT/HVRT) with reactive current injection.
- Reactive capability and voltage control at the POI.
- Active power control, ramp rates, curtailment response.
- Frequency response (LFSM-O/U, FSM).
- Power quality: harmonics, flicker, unbalance.
- Protection settings approved; SCADA and metering.
- Validated models; compliance test programme; witness by MEWRE.

Energisation hold points (in order)

1. Protection settings approved and tested
2. SCADA point-to-point checks
3. Telecom / teleprotection
4. Metering sealed
5. Switching programme approved
6. Back-feed HV, soak
7. MV energised progressively
8. Inverters / PPC
9. Witnessed compliance tests
10. Trial (reliability) run
11. COD

> **HOOK** "Protect, see, talk, count, plan; then power up from the top down, test, run, declare."

SCADA, communications, cyber

- Field → plant ring → SCADA / PPC / historian → gateway (**IEC 60870-5-104**) → OPGW fibre → national control centre EMS.
- **IEC 61850** inside substations: MMS (data), **GOOSE** (fast trip messages, ~4 ms), SV (sampled values); SCD file under change control.
- Export-line protection: 87L main, 21 back-up.
- **IEC 62443**: zones and conduits, security levels, DMZ, jump host + MFA, no rogue 4G modems.

Questions that show you understand grid connection

ASK Which grid-code version applies to the new PV and BESS plants, and who runs the compliance tests?

ASK What is the minimum fault level at the POI and the resulting SCR? Has an EMT study been done with the vendor's real models?

ASK Is reactive capability at 50 °C met by inverters alone or with a STATCOM?

ASK How are the 400 kV works coordinated with the plant's COD? (MEWRE's grid works are the IPP's critical path and a deemed-energy risk.)

ASK Who approves protection settings, and when is the switching programme issued?

9 · Solar PV

module 2, trial Day 2

Sunlight as fuel

Irradiance	Power of sunlight now, **W/m²** (≈ 1,000 at clear noon).
Irradiation	Energy over time, **kWh/m²** per day or year.
GHI	Global horizontal: all light on a flat surface. What fixed PV uses.
DNI	Direct normal: the beam from the sun's disc. What **CSP** needs (mirrors can't focus diffuse light).
DHI	Diffuse horizontal: scattered sky light (high in dust).
POA	Plane-of-array irradiance: what the tilted module actually receives.

$GHI = DNI \cdot \cos\theta_z + DHI$ θ_z = sun's zenith angle

Kuwait (29.3° N): high sun, extreme heat (record 53.9 °C, Mitribah 2016), frequent dust and shamal winds. Annual GHI roughly 1,900 to 2,100 kWh/m² V.

How a cell works

- Photons free electrons in silicon; the **p-n junction** pushes them one way → **DC**.
- **I-V curve**: from I_{sc} (short-circuit current) to V_{oc} (open-circuit voltage); the knee is the **maximum power point (MPP)**. Inverters track it (MPPT).
- **Heat lowers voltage** and power: about −0.3 to −0.4 %/°C (PERC higher loss, TOPCon and HJT lower).
- Module types: mono PERC, **TOPCon**, HJT; **bifacial** modules also use light reflected from the ground (+5 to 10 %).

$P = P_{STC} \cdot [1 + \gamma\,(T_{cell} - 25\,°C)]$ $\gamma \approx -0.0035/°C$

$T_{cell} \approx T_{amb} + (NOCT - 20)/800 \cdot G$

Kuwait July: ambient 45 °C, cell ~70 °C → ~16 % less power than the label.

STC	Standard test conditions: 1,000 W/m², 25 °C cell, AM 1.5. Module "Wp" is at STC.

Plant anatomy

Modules → Strings (≤ 1,500 V DC) → Combiner → Inverter (DC→AC) → MV transformer → 33 kV collector → Plant substation → 400 kV

MWp vs MWac	DC module capacity vs AC inverter export limit. Zone 1 base case: 1,430 MWp / 1,100 MWac (DC/AC 1.3, illustrative).
DC/AC ratio	More DC than AC (1.2 to 1.4) to fill the inverter in mornings, evenings and heat; costs some **clipping** at noon.
Central vs string inverters	Few big units (cheap, single points of failure) vs many small ones (redundant, more units to maintain).
Single-axis tracker	Rows rotate east to west: +15 to 25 % yield; **backtracking** avoids row shading; **stow** (flat) in high wind.
GCR	Ground coverage ratio: module width ÷ row pitch (~0.3 to 0.45).

TRAP Quoting MWp as if it were MW at the peak hour. Always say MWac, and the evening contribution is near zero.

Losses, yield and PR

Specific yield	kWh per kWp per year (base case 2,100 kWh/kWp, illustrative).
PR	**Performance ratio**: actual output ÷ what the irradiance allowed, independent of weather. Good plant ~80 to 85 %; temperature-corrected PR used in guarantees.
Soiling	Dust on modules: ~0.3 to 0.5 %/day in Kuwait-type climates; cleaning restores. Dew + dust can cement.
Degradation	~0.4 to 0.5 %/yr after first-year loss (LID/LeTID).
Loss waterfall	Irradiance → soiling → temperature → mismatch → DC cables → inverter → clipping → AC/transformer → availability → export.

$PR = Y_f / Y_r$, $Y_f = E_{AC}/P_{DC}$ (h), $Y_r = H_{POA}$ / 1 kW/m² (h)

Example: 2,665 MWh from 50 MWp in a month = 53.3 h; POA 65 kWh/m² = 65 h → PR = 82 %.

Optimal cleaning interval $N^* = \sqrt{(730\ C / (s\ E\ p))}$ C cost per clean, s soiling rate/day, E annual energy, p price

P50 / P90

P50	Median yearly yield: exceeded 1 year in 2.
P90	Exceeded 9 years in 10: lenders size debt on it. More measured site data → lower uncertainty → higher P90 → more debt → lower tariff.

$P90 = P50 \cdot (1 - 1.282\ \sigma)$ σ = combined uncertainty, ~6 to 8 %

Field QA and what a PM checks

- Factory audits and flash tests; EL (electroluminescence) imaging for cracks.
- Thermal imaging for hot spots; string I-V curve tracing at commissioning.
- Tracker commissioning: angles, backtracking, stow function.
- PR test: temperature-corrected, agreed method, duration, data filters.

ASK What is the guaranteed PR and how is it corrected for temperature and soiling? Which cleaning regime is assumed in the yield model?

10 · CSP, wind and batteries

modules 3, 4, 5; trial Day 2

Concentrated solar power (CSP)

Principle	Mirrors focus **DNI** → heat a fluid → steam turbine (like a thermal plant, sun instead of fuel).
Parabolic trough	Curved mirror rows heat a receiver tube (thermal oil, ~390 °C). Shagaya CSP is a trough plant.
Tower	Heliostat field focuses on a central receiver; molten salt ~565 °C; higher efficiency.
Thermal storage	Hot and cold **molten-salt tanks** run the turbine after sunset.

$E_{th} = m \cdot c_p \cdot \Delta T$ storage hours = $MWh_{th} \cdot \eta_{cycle} / MW_e$

Shagaya Phase I: 50 MW CSP of the 70 MW H; 225,000 MWh/yr figure → CF 51 % if all from CSP (37 % across 70 MW). Storage of about 9 h is an estimate (1,200 MWh_{th} × 0.37 ÷ 50 MW). Performance since 2019 not public L.

- Dust hurts CSP more than PV: a fixed receiver loss turns 10 % less DNI into 12 to 15 % less electricity.
- **CSP vs PV + BESS:** PV + BESS usually cheaper for 2 to 4 h; CSP's niche is long storage (8 h+) and grid support from a synchronous turbine (inertia, fault current).

Wind

$P = ½\ \rho\ A\ v^3\ C_p$ Betz limit $C_{p,max} = 16/27 \approx 0.593$

- **Speed cubed:** 10 % more wind → 33 % more power. Mean of the cubes ≫ cube of the mean (Weibull distribution).
- Hot air is thinner (ρ falls ~10 % from 15 to 45 °C) → less power.
- Shagaya Phase I wind: 10 MW H; mean ~7.9 m/s cited, figures conflict V.
- Desert issues: sand erosion of blade leading edges, heat derating of converters and gearboxes.

Battery storage (BESS) basics

MW	Power: how fast it charges or discharges (the tap).
MWh	Energy: how much it holds (the tank). Duration h = MWh/MW.
C-rate	P/E: 0.25C = 4-hour battery.
RTE	Round-trip efficiency, ~85 to 88 % at the connection point.
DoD / SoC	Depth of discharge / state of charge; usable window e.g. 2.5 to 97.5 % SoC.
LFP vs NMC	LFP (lithium iron phosphate): safer, longer life, standard for grid storage; NMC: denser, higher fire risk.
PCS / BMS / EMS	Power conversion system (inverter), battery management (cell protection), energy management (dispatch).

Energy chain example: DC nameplate 1,003 MWh → 953 → 929 → **905 MWh at the meter**: specify which MWh is guaranteed.

Kuwait plan: 1 to 1.5 GW / 4 to 6 GWh; first 500 MW for summer 2027 M-L. Global turnkey cost ~USD 165/kWh (BNEF 2024, 4 h).

Battery life and safety

- **Degradation:** capacity fades with cycles, time and **heat** (aging roughly doubles per +10 °C).
- **Augmentation:** add modules later to keep the guaranteed MWh, or accept LDs.
- **Warranty** limits: cycles per day, temperature, SoC window; exceeding them voids it.
- **Thermal runaway:** cell overheats → venting flammable gas → fire spreading cell to cell. Mitigation: BMS, spacing, gas detection, ventilation, deflagration vents, water and suppression; tested to **UL 9540A**; design to **NFPA 855**.
- Desert HVAC: cooling energy and its availability are critical; who is responsible for HVAC in the O&M contract.

Guaranteed energy $E_g(y) = E_{BOL} \cdot (1 - d)^y$

ASK Is the BESS PCS grid-forming capable? Which MWh is guaranteed (DC or at the POI), at what temperature and cycling?

11 · Project delivery and control

modules 7, 8; trial Day 3

Lifecycle and gates

Idea → Feasibility → Permits / EIA → Design → Procurement → Construction → Testing & commissioning → COD → Taking over → Defects period → O&M

Stage gate: review against written criteria → Go / Go with conditions / Recycle / Stop. Funding released stage by stage.

AACE class	Accuracy	When
5	−50 / +100 %	idea
4	−30 / +50 %	feasibility
3	−20 / +30 %	budget authorisation
2	−15 / +20 %	control / bid
1	−10 / +15 %	check estimate

Design and the owner's engineer

Concept → basic (FEED) → detailed → IFC	Design matures; IFC = issued for construction.
Owner's engineer (OE)	Consultant who reviews design, witnesses tests, checks progress for MEWRE. **Advises; MEWRE decides.**
FIDIC Engineer	Administers the contract (certifies, determines claims): must act neutrally on determinations.
Lenders' technical advisor	Works for the banks on IPPs.
Review codes	A: proceed · B: proceed with comments, resubmit · C: revise, don't proceed · D: information only. Typical review period 21 days; late review may be deemed no-objection.

Testing, commissioning, COD

FAT (factory) → SAT (site) → Cold commissioning → Hot commissioning / energisation → Grid compliance tests → Performance (PR) test → Reliability run → COD → Taking-Over Certificate

COD	Commercial operation date: energy is paid from here; delay damages stop.
Punch list	Category A: must close before taking over. Category B: minor, close during defects period.
Taking-Over Certificate (TOC)	Care and risk pass to the employer; delay LDs stop; defects period starts; half retention released.
DNP	Defects notification period, 12 to 24 months (FIDIC default 365 days, extendable up to 2 years).
Performance Certificate	Within 28 days after the last DNP ends: contract complete.

Scope and WBS

- **WBS**: break the scope into deliverables (e.g. PV blocks, substation, 400 kV line, SCADA, civil); work packages at the bottom.
- **WBS dictionary**: what each element includes and excludes.
- **Interfaces**: physical, functional, contractual, organisational; MEWRE grid works vs IPP scope at the battery limit (POI).
- **Scope creep**: small accepted changes adding up; control through change requests.

Schedule and critical path (CPM)

Critical path	Longest chain of dependent activities: zero float; it sets the end date.
Total float	How far an activity can slip without moving the end date.
Free float	Slip allowed without delaying the next activity.
Baseline	Approved schedule to measure against. **Data date**: progress cut-off.
P6	Primavera P6, standard scheduling software.

$$TF = LS - ES = LF - EF \quad \text{delay to end} = \max(0, \text{slip} - TF)$$

Slip of 12 weeks on an activity with 8 weeks float → 4 weeks late finish. FIDIC 8.3: programme within 28 days, Engineer reviews in 21 days.

Merge bias: two parallel paths each 50 % likely on time → only 25 % that both are.

Earned value (EVM)

PV	Planned value: budgeted cost of work scheduled.
EV	Earned value: budgeted cost of work done.
AC	Actual cost of work done.
BAC	Budget at completion.

$$CV = EV - AC \quad SV = EV - PV \quad CPI = EV/AC \quad SPI = EV/PV$$

$$EAC = BAC/CPI \ \text{(or } AC + (BAC - EV)) \quad ETC = EAC - AC \quad VAC = BAC - EAC$$

$$TCPI = (BAC - EV)/(BAC - AC)$$

BAC 100, PV 40, EV 36, AC 45 → CPI 0.80, SPI 0.90, EAC 109 to 125, TCPI 1.16.

> **HOOK** EV is "what we earned", compare it to what we planned (SPI) and what we spent (CPI). Below 1 = behind or over.

Risk, uncertainty, change

$$\text{Risk score} = P \times I \quad EMV = \Sigma\, p_i \cdot I_i \quad \text{contingency} \geq EMV, \text{ budget at P80}$$

$$\text{PERT } t_e = (a + 4m + b)/6 \quad \sigma = (b - a)/6 \quad P80 \approx \Sigma\, t_e + 0.84\, \sqrt{(\Sigma\, \sigma^2)}$$

- **Responses:** avoid, transfer, mitigate, accept (threats); exploit, share, enhance (opportunities).
- **Monte Carlo:** thousands of simulated schedules → P50 / P80 dates.
- **Change control:** request → impact (cost, time, risk) → approval at the right authority → contract instruction → baseline update.

Monthly report (page 1)

- RAG status per area (schedule, cost, HSE, quality, risks, approvals) with trend and the criterion used.
- Every red row ends in a **decision or ask**.
- Key dates vs baseline; critical path; top 5 risks with owners; decisions needed.
- HSE and QA/QC in one line each; detail in appendices.

12 · Contracts (FIDIC 2017)

module 9, trial Day 4

The FIDIC books

Book	Design	Price	Administrator	Ground risk
Red (Construction)	Employer	Remeasured BoQ	Engineer	Employer
Yellow (Plant & Design-Build)	Contractor	Lump sum	Engineer	Employer
Silver (EPC/Turnkey)	Contractor	Fixed lump sum	Employer's Rep.	Contractor
Gold (DBO) / Emerald (tunnels)	Contractor	Lump sum + operation	ER / Engineer	Contractor-leaning / shared

HOOK Red = "Read our drawings"; Yellow = "You design"; Silver = "Single price, Seller takes the risk".

Risk principle: give each risk to the party that controls it, otherwise the cheapest bearer. Shifting risk costs a premium RP = k · EV with k > 1.

Parties: Employer (MEWRE), Engineer / Employer's Representative, Contractor, subcontractors, DAAB (dispute board).

Time limits to memorise

Notice of Claim after awareness (20.2.1)	**28 days** (time-bar)
Fully detailed claim (20.2.4)	**84 days**
Performance security after LoA (4.2)	28 days
Programme after commencement / Engineer review (8.3)	28 / 21 days
Agreement then determination (3.7)	42 + 42 days
NOD on determination or DAAB decision	28 days
DAAB decision	84 days, binding at once
IPC issue / payment (14.6, 14.7)	28 / 56 days
Taking-Over Certificate (10.1)	28 days after application
Performance Certificate (11.9)	28 days after last DNP
Base Date	28 days before tender deadline

TRAP Minutes are not Notices. A claim or instruction still needs the contractual Notice; the Employer's own delay-damage claims need a 28-day notice too.

Variations, claims, delay

Variation	Instructed change; valued by contract rates → derived rates → cost plus profit (default 5 %).
Claim	Request for time and/or money under a sub-clause; needs notice + substantiation.
EOT	Extension of time on the updated programme; float shared unless the contract says otherwise.
TIA	Time impact analysis: insert the event in the contemporaneous programme (preferred method, SCL Protocol).
Concurrency	Employer and contractor delays together: **time, not money**.
Disputes	Engineer's determination → DAAB → NOD in 28 days → amicable settlement → arbitration. Kuwait forum: verify with legal.

$$\text{EOT} = \max(0,\ \text{delay} - \text{total float})$$

Securities

Security	Typical	Released
Bid bond	1 to 5 %	Signature / award to others
Performance security	10 %	Performance Certificate (often 5 % after TOC)
Advance payment guarantee	= advance	Reduces with recovery
Retention	5 % of contract	½ at TOC, ½ after DNP
Warranty bond	2.5 to 5 %	End of DNP
Parent company guarantee	Full liability	All obligations end

Advance recovery starts when certified payments exceed 10 % of the contract; recovered at 25 % of each payment certificate.

IN PRACTICE Track bond expiries 60 days ahead: a bond expiring before the Performance Certificate is a gap. On-demand bonds pay on a compliant demand (URDG 758), but the employer may claim only what it is entitled to.

Liquidated damages (LDs)

Delay LDs	Agreed amount per day late after EOT; sanity 0.03 to 0.1 % of contract price per day; cap 10 to 20 %.
Performance LDs	Buy-down for missing PR or capacity guarantees; cap 10 to 15 %; aggregate cap 20 to 30 %.
Minimum performance	Below it: remedy or reject, no buy-down.
Penalty test	Common law (Makdessi): legitimate interest, not out of proportion. Civil law: الشرط الجزائي, court may reduce (Kuwait articles: verify).

$$\text{Total delay LDs} = \min(\text{LD}_d \times \text{days late}, \text{cap} \times \text{contract price})$$

$$\text{PR buy-down} = (\text{PR}_g - \text{PR}_m)/\text{PR}_g \cdot E_{yr} \cdot T \cdot AF, \quad AF = [1 - (1+i)^{-n}]/i$$

KWD 20 m EPC, 0.1 %/day, 10 % cap → KWD 20k/day, capped at KWD 2 m (100 days).

IPP / PPA structure

Sponsors (equity) + Lenders (debt) → SPV (project company) → EPC + O&M contracts ⇄ PPA with MEWRE (offtaker)

PPA	Power purchase agreement: solar 30 years with MEWRE; energy-only tariff for PV; capacity + energy for thermal and BESS.
Take-or-pay / deemed energy	Offtaker pays for energy it could not take because of grid outage, curtailment or late grid connection.
Termination payments	SPV default (≤ debt) < natural force majeure (debt + equity) < offtaker default (debt + equity + return).
Direct agreement	Lenders' step-in, cure periods, novation.

$$\text{Revenue } R = T \cdot (E_{delivered} + E_{deemed}) \quad \text{capacity + energy: } R = CC \cdot P \cdot \min(1, A/A_{ref}) + EC \cdot E$$

Not confirmed for Kuwait: PPA currency, government support, termination regime, deemed-energy terms V.

IN PRACTICE MEWRE's grid works (400 kV line, substation bay) are on the IPP's critical path: late grid = deemed energy paid by MEWRE.

13 · Money: NPV, IRR, LCOE, project finance

module 10, trial Day 4

Time value of money

$$PV = CF_t/(1+r)^t \quad NPV = \Sigma\, CF_t/(1+r)^t$$

$$\text{IRR: the r where NPV} = 0 \quad \text{Payback} = Y + |C_Y|/CF_{Y+1}$$

$$WACC = (E/V)\, k_e + (D/V)\, k_d\, (1 - T)$$

$$\text{Annuity } PV = A \cdot [1 - (1+r)^{-N}]/r \quad (1+r_n) = (1+r_r)(1+i)$$

- **Decide** on NPV; **communicate** with IRR (traps: multiple IRRs, scale).
- Base WACC: 0.2 × 9 % + 0.8 × 5.5 % = **6.2 %**. Discount factors at 6.2 %: year 10 = 0.548, 20 = 0.300, 30 = 0.165.
- Units: 1 KWD ≈ 3.25 USD; 1 US¢/kWh = 3.08 fils/kWh.

LCOE

$$LCOE = \Sigma\, (I_t + O_t)(1+r)^{-t} / \Sigma\, E_t(1+r)^{-t}$$

$$CRF = r(1+r)^n / [(1+r)^n - 1] \quad \text{quick LCOE} \approx (I_0 \cdot CRF + O)/E$$

Base case (illustrative): 1,100 MWac / 1,430 MWp, 2,100 kWh/kWp, 3,003 GWh/yr, CF 31.2 %, capex USD 560 m (0.39 USD/Wp), opex USD 7.5 m/yr, 30 years → **LCOE 1.764 US¢/kWh at 6.2 %**.

Driver (tornado)	Swing (US¢/kWh)
Yield ±10 %	0.356
Capex ±10 %	0.288
WACC ±1 pt	0.288
Opex ±20 %	0.129

Gulf records (reported): MBR 5 ≈ 1.69 (2019), Al Dhafra ≈ 1.35 (2020), Sudair ≈ 1.24 (2021), Al Shuaibah ≈ 1.04 US¢/kWh (2021). Kuwait Zone 1 tariff: not published.

Project finance

CFADS	Cash flow available for debt service = revenue − opex − tax − Δworking capital − reserves.
DSCR	CFADS ÷ (interest + principal) each period; lenders size debt at ~1.2 to 1.3×; default ~1.05×; dividend lock-up 1.10 to 1.20×.
LLCR / PLCR	PV of CFADS over loan life / project life ÷ debt.
Gearing	Debt share, Gulf solar 75 to 85 %.
DSRA	Debt service reserve account, ~6 months.

$$DSCR = CFADS/(I + P) \quad D_{max} = \Sigma\ (CFADS_t/DSCR)/(1 + k_d)^t$$

Waterfall: opex and tax → senior debt → DSRA → maintenance reserve → lock-up test → dividends.

Base case at 1.80 US¢: debt 448, equity 112 (USD m), DSCR 1.287×, project IRR 6.45 %, equity IRR 8.97 %; P90 year-1 DSCR 1.183×.

Comparing tariff bids

$$\text{Levelised tariff } LT = \Sigma\ T_t\ E_t\ (1+d)^{-t} / \Sigma\ E_t\ (1+d)^{-t}$$

- Publish the discount rate, energy profile and indexation **before** bids; a higher d favours back-loaded (escalating) bids.
- 3.90 fils escalating 1.5 %/yr levelises to 4.54 fils at 7 %, above a flat 4.20.
- An abnormally low bid (−20 % vs median) can mean no financial close, delay, quality cuts and claims: test it.
- Two-envelope (KAPP): technical first, then financial. QCBS for consultants: $S = w_T\ S_T + w_F\ S_F$, $S_F = 100\ P_{min}/P_i$; fix weights before opening.

Budget and cash flow

- Cycle: budget → commitment → accrual (certified IPC) → expenditure.
- Fiscal year 1 Apr to 31 Mar: phase cash needs by FY (e.g. KWD 45 m, 24 months from Jun 2026 → FY26/27 14.3, FY27/28 26.8, FY28/29 2.8 m).
- S-curve: $F(t) = (1 - \cos(\pi t/T))/2$.

14 · Procurement in Kuwait · 15 · Operations and warranties

modules 11, 12

Procurement methods

Public tender	Default; open to all registered bidders. CAPT default award: lowest technically compliant price.
Limited tender	Invited, prequalified bidders.
Practice (ممارسة)	Negotiation with a few suppliers.
Direct order	Below the direct-contracting limit or justified exceptions (Law 94/2026: up to KD 1 m per FY).
Clarifications	In writing, to all bidders, no change to price or substance.

- **Anti-splitting:** all orders for the same item in a FY count together.
- **Local content** ≥ 30 % (define the denominator in the contract); national product preference $P_{nat} \le (1 + \pi)\ P_{min}$.
- **Contracting without approval** is a violation (28 cases worth KD 63 m reported).

Delays and how to manage them lawfully

Evidence: Zone 1 bid deadline moved Sep 2025 → Jan 2026; Al-Khairan 3 extensions; Dibdibah and Abdaliya cancelled.

- Map the approval calendar at the start of the year.
- Pre-cleared, complete files; standard documents.
- Prepare the next gate in parallel.
- Settle bankability before the RFP; manage bid validity.
- Escalate with the **KWD/day** cost of delay; keep a referral register.
- Good Arabic in official letters. **Never start before approval.**

IN PRACTICE Attack the approval step with the biggest variance first (PERT): that is where the schedule risk lives.

Plant KPIs

Time availability	Share of hours the plant could run (capacity-weighted).
Energy availability	Actual ÷ (actual + energy lost to outages): an outage at noon in July counts far more than at dawn in January.
PR	Hit by both outages and soiling; availability is hit only by outages: read them together.
Loss signatures	Flat top at 100 % = clipping; plateau at a set-point = curtailment; step = string or inverter failure; slow decline = soiling.

$$A_E = E_{act}/(E_{act} + E_{lost}) \quad malus = (A_g - A_E)(E_{act} + E_{lost})\ T$$

Availability guarantee 98.5 to 99.5 %; malus cap 50 to 100 % of the O&M fee.

O&M contracts, spares, warranties

- **O&M scope:** monitoring, preventive and corrective maintenance, cleaning, spares, reporting; HVAC for BESS.
- **Spares:** size stock with a Poisson model (mean 9.6 failures/yr: 10 spares → 63 %, 15 → 96 % service level).
- **Module warranties:** product ~10 to 12 years; performance 25 to 30 years (linear, e.g. −0.4 to −0.55 %/yr).
- **Claims:** prove with measured data (I-V curves, EL images) against the warranty threshold.
- **Handover:** as-builts, settings, SCADA credentials, spares, warranties, DNP interface agreement.

16 · Working in the ministry

modules 13, 14; trial Day 4

Writing: the decision memo (1 page, ~450 words)

1. **Decision required** (+ deadline + recommendation): BLUF, bottom line up front
2. Issue
3. Background (with references)
4. Options, including do-nothing (cost / time / risk)
5. Recommendation and the decisive reason
6. Risks
7. Decision line
8. Attachments (draft letter ready to sign)

Seniors read ~200 words/min: one page ≈ 2 minutes. MW as a share of the 17,540 MW peak; KWD as a share of budget or contingency.

Arabic official letter (top to bottom)

1. الرقم / التاريخ
2. السيد/ ... المحترم
3. الموضوع
4. تحية طيبة وبعد،،،
5. بالإشارة إلى كتابكم رقم ... المؤرخ في ... بشأن ...
6. نفيدكم بما يلي (numbered points)
7. يرجى التكرم بـ ...
8. وتفضلوا بقبول فائق الاحترام والتقدير،،،
9. Signature, المرفقات / نسخة إلى

Valid contractual notice: "NOTICE / إشعار" in the subject; exact sub-clause of *your* contract; right sender and address; within the time limit; facts not adjectives; تحتفظ الوزارة بكافة حقوقها بموجب العقد.

Presenting and bad news

- **Pyramid:** answer first → 3 supporting points → evidence. Opener: Situation, Complication, Question, Answer.
- **Three slides:** the answer, why, the asks. Action titles; appendix = answer bank.
- Date slips as a range; every risk with an owner and "when we'll know more".
- **Bad news:** cause → impact → plan → ask. Early beats perfect; never surprise your superior.
- Never guess numbers: "I'll confirm by [date]". Never contradict your superior in front of externals. Commitments only in signed letters.

Meetings, consultants, negotiation

- **Progress meeting agenda:** safety → previous actions → progress vs programme → engineering and procurement → quality → risks → commercial → decisions → new actions (owner + date read back).
- Minutes within 3 working days; comments within 5. **Minutes are evidence**: rebut disputed statements on the record.
- **Escalation ladder:** discussion → minuted action → letter → Notice / NCR → contractual remedies → senior escalation → termination.
- **Principal-agent:** the OE and contractor know more than you; control with scope, incentives, reporting, **verification**, records.
- **Negotiation on public money:** settle on entitlement + substantiated quantum, not split-the-difference. Interests not positions; objective criteria; know your **BATNA** (best alternative) and the **ZOPA** (zone of possible agreement).

Using AI tools safely

- Good at: drafting, summarising, structuring, translating, checking consistency. Bad at: facts it was not given, recent laws, article numbers, arithmetic without checking.
- **Hallucination:** confident invented facts. Verify every law, article, number against the primary source.
- **Never paste** ministry documents, bidder data or personal information into unapproved public tools; use approved enterprise tools only.
- Trust matrix: your own supplied text + low stakes → light check; model memory + legal effect → full verification.

17 · Trial-week playbook

use on the day

Day by day

- **Day 1, listen and map:** names, roles, projects; notes in minutes format (decision, action, owner, date).
- **Day 2, offer something small:** a written meeting summary within 24 h; ask which documents you may read.
- **Day 3, show judgement:** on any issue, ask what the contract or approval chain says before opining.
- **Day 4, produce:** one useful page: pipeline tracker, risk list or bilingual brief.
- **Day 5, close well:** thank people by name; written summary of what you learned and your first 90 days.

Questions that show you prepared (2 or 3 a day)

1. Which projects is the team working on this quarter, and which decision is next for each?
2. For Zone 1, what is the next milestone: PPA signature, financial close or construction start?
3. Where does the ministry sit in the approval chain for this project, and what is the file waiting on?
4. Which grid-code version applies to new PV and BESS, and who runs compliance tests?
5. Is BESS procured inside the PV IPPs or as a separate tender?
6. Who is the Owner's Engineer, and what do you rely on them for?
7. How is status reported to leadership?
8. What caused the longest recent delays, and what unblocked them?
9. Which risks worry the team most: grid connection, land, approvals, supply chain, heat?
10. How are 400 kV works coordinated with solar COD dates?
11. Is there a standard template for briefs and letters?
12. What would a useful contribution from me look like by the end of this week?

What hurts credibility

- Quoting MWp as MW at the peak hour.
- Calling a G2G or framework deal "contracted" before signature.
- A number without its source or date.
- Criticising past decisions before knowing their approval history.
- Pasting ministry documents or bidder data into public AI tools.
- Promising dates on behalf of KAPP, CAPT or the Audit Bureau.

Your 60-second introduction (fill in, practise aloud)

"I'm [name], a systems and energy engineer. For the past [n] years I've worked on [background, e.g. automation and control of large systems, including battery and hybrid controls]. I've spent the last weeks preparing specifically for MEWRE's renewable programme: the Shagaya and Zone 1 and 2 PV projects, the BESS plans, and how projects move through KAPP, CAPT and the Audit Bureau. What I bring is [one strength, e.g. systems thinking and turning complex technical issues into clear decisions]. This week I'd like to understand how the team works and leave you with something useful in writing."

18 · Acronyms and terms A to Z

AACE	Cost estimate classes 1 to 5
AC / DC	Alternating / direct current
AC (EVM)	Actual cost
AGC	Automatic generation control (secondary)
AIS / GIS	Air / gas-insulated switchgear
APG	Advance payment guarantee
BATNA	Best alternative to a negotiated agreement
BESS	Battery energy storage system
BIL	Basic lightning impulse insulation level
BLUF	Bottom line up front
BMS / EMS	Battery / energy management system
BoQ	Bill of quantities
CAPT	Central Agency for Public Tenders
CB	Circuit breaker
CF	Capacity factor
CFADS	Cash flow available for debt service
COD	Commercial operation date
CPI / SPI	Cost / schedule performance index
CPM	Critical path method
CRF	Capital recovery factor
CSP	Concentrated solar power
CT / VT / CVT	Current / voltage / capacitive voltage transformer
DAAB	Dispute avoidance and adjudication board
DHI / DNI / GHI	Diffuse / direct normal / global horizontal irradiance
DNP	Defects notification period
DoD / SoC	Depth of discharge / state of charge
DSCR	Debt service coverage ratio
DSRA	Debt service reserve account
EAC / ETC / VAC	Estimate at / to complete; variance at completion
ECWPA	Energy conversion and water purchase agreement
EIA	Environmental impact assessment
EMT	Electromagnetic transient (study/simulation)
EMV	Expected monetary value
EOT	Extension of time
EPC	Engineering, procurement, construction
ESCR / WSCR	Effective / weighted short-circuit ratio
EV / PV (EVM)	Earned / planned value
FAT / SAT	Factory / site acceptance test
FEED	Front-end engineering design
FFR	Fast frequency response
FIDIC	International Federation of Consulting Engineers (contract suites)
FRT / LVRT / HVRT	Fault / low / high voltage ride-through
FSM / LFSM-O/U	Frequency sensitive mode / limited, over / under
G2G	Government to government
GCCIA	GCC Interconnection Authority
GCR	Ground coverage ratio
GFL / GFM	Grid-following / grid-forming inverter
GOOSE	Fast IEC 61850 trip messaging
GSU	Generator step-up transformer
HVDC	High-voltage direct current
IBR	Inverter-based resource
IED	Intelligent electronic device (protection relay)
IFC	Issued for construction
IPC	Interim payment certificate
RoCoF	Rate of change of frequency

IPP / IWPP	Independent power / water and power producer
IRR / NPV	Internal rate of return / net present value
KAPP	Kuwait Authority for Partnership Projects
KISR	Kuwait Institute for Scientific Research
LCOE	Levelised cost of electricity
LDs	Liquidated damages
LFP / NMC	Lithium iron phosphate / nickel manganese cobalt
LLCR / PLCR	Loan / project life coverage ratio
LoA	Letter of acceptance
MPPT	Maximum power point tracking
MWp / MWac	DC module / AC export capacity
N-1	System withstands loss of any one element
NCR	Non-conformance report
NOD	Notice of dissatisfaction
OE	Owner's engineer
OHL	Overhead line
OLTC	On-load tap changer
ONAN / ONAF	Transformer cooling: natural / forced air
OPGW	Optical ground wire (fibre in the earth wire)
P50 / P90	Yield exceeded 50 % / 90 % of years
PCS	Power conversion system
PERT	3-point estimate (a + 4m + b)/6
PF	Power factor
PLL	Phase-locked loop
POA	Plane-of-array irradiance
POI / PCC	Point of interconnection / common coupling
PPA	Power purchase agreement
PPC	Power plant controller
PPP	Public-private partnership
pu	Per unit
PR	Performance ratio
QCBS	Quality and cost based selection
RAG	Red / amber / green status
RfG	EU grid code "Requirements for Generators"
RFQ / RFP	Request for qualification / proposal
RMS	Root mean square; also phasor stability study
RMU	Ring main unit
RoCoF	Rate of change of frequency
RTE	Round-trip efficiency
SCADA	Supervisory control and data acquisition
SCR	Short-circuit ratio
SIL	Surge impedance loading
SPV	Special purpose vehicle (project company)
STATCOM / SVC	Static synchronous compensator / static var compensator
STC / NOCT	Standard test / nominal operating cell temperature
TF / FF	Total / free float
THD	Total harmonic distortion
TIA	Time impact analysis
TOC	Taking-over certificate
TOV	Temporary overvoltage
UFLS	Under-frequency load shedding
Um	Highest voltage for equipment
WACC	Weighted average cost of capital
WBS	Work breakdown structure
X/R	Reactance-to-resistance ratio
ZOPA	Zone of possible agreement